

Pulse-Agile Waveform and STAP Filter Design for Airborne MIMO DFRC Systems

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Abstract—Moving-platform dual-function radar and communication (DFRC) systems have attracted increasing attention in recent years due to their flexibility in deployment and wide-area coverage capabilities. As a critical intermediate layer in integrated space-air-ground networks, airborne DFRC systems are of great research significance and typically rely on space-time adaptive processing (STAP) to enable target sensing in strong clutter environments. However, the pulse-repetitive transmission required by STAP severely limits the achievable communication rate. To enable robust target detection in clutter-dominated environments while simultaneously ensuring high-rate communication and strong jamming resilience, we propose a method that employs pulse-agile transmit waveforms at the transmit side to enhance communication throughput and improve resilience against repeater jamming, while the receive filters are designed to enable coherent target integration in the desired direction and robust clutter suppression, thereby improving detection performance. The resulting optimization problem is efficiently solved within an alternating direction method of multipliers framework. Simulation results demonstrate the effectiveness and advantages of the proposed method in both sensing and communication.

Index Terms—dual function radar-communication, space-time adaptive processing, inter-pulse waveform agility, optimization

I. INTRODUCTION

With the rapid evolution of 6G communication technologies, integrated sensing and communication (ISAC), as one of its core enablers, has attracted significant attention due to its tremendous application potential in smart cities, low-altitude economy, and the Internet of Thing [1], [2], [3]. At its core, ISAC employs a unified hardware platform to simultaneously enable sensing and communication over a shared frequency band, thereby achieving substantially higher overall efficiency. However, most existing ISAC studies primarily focus on augmenting terrestrial base stations with sensing capabilities (e.g., [4], [5], [6], [7], [8]). In recent years, mobile platforms have attracted growing attention due to their flexible deployment and extended sensing coverage. Existing studies on DFRC systems for mobile platforms mainly focus on endowing communication satellites with wide-area sensing capabilities or enabling fine-grained local sensing using UAVs equipped with small-scale arrays (e.g., [9], [10], [11], [12], [13], [14], [15]). Beyond these platforms, airborne DFRC systems offer distinct advantages: compared with UAVs, they support heavier payloads and higher operating altitudes, enabling wider-area sensing and multiuser communications; compared with satellites, they provide more flexible deployment and can de-

liver all-weather, high-resolution sensing and communication services over hotspot regions.

Airborne DFRC systems operate in a down-looking mode and receive ground echoes (clutter) from a wide range of azimuth angles during sensing operations, which can severely mask the target returns. Space-time adaptive processing (STAP), leveraging its joint processing capability across both spatial and temporal dimensions, is a key technology for achieving reliable target detection on moving platforms operating in strong clutter environments [16], [17], [18], [19]. Specifically, STAP exploits multi-antenna and multi-pulse data to separate clutter from targets in the space-time domain, thereby enabling effective clutter suppression while preserving target returns. The fast-slow time signal processing structure of STAP provides a natural framework for embedding communication information. However, conventional STAP [20] relies on repeatedly transmitting identical pulses to capture target Doppler information, which severely limits the achievable communication rate. This discrepancy necessitates the development of specialized space-time signal processing techniques tailored for time-varying pulse structures.

Motivated by these considerations, this paper investigates a joint transmit waveform and receive filter design framework based on inter-pulse waveform-agile STAP. At the transmitter, the integrated waveform is modulated with communication information and varies in both the fast-time and slow-time domains to enhance the communication rate, while transmit beamforming is employed to ensure effective energy focusing. At the receiver, a dedicated receive filter is designed to guarantee high-quality coherent integration, which is essential for preserving sensing performance under time-varying waveforms. Furthermore, we impose an inter-pulse waveform similarity constraint to enhance the resistance against repeater jamming. Based on the above, a multivariate optimization problem with multiple constraints is formulated, and an efficient ADMM-based solution is developed. Simulation results demonstrate the effectiveness of the proposed approach in terms of both sensing and communication performance.

II. MODELING

We consider an airborne multiple-input multiple-output (MIMO) DFRC system with a sidelooking antenna array comprising N_T transmit and N_R receive elements, which simultaneously performs target sensing and serves K down-

link users. The radar and communication signal models are presented as follows.

A. Radar Model

As mentioned earlier, the transmit waveform varies from pulse to pulse. Therefore, we denote the transmit waveform of the m -th pulse as

$$\mathbf{X}_m = [\mathbf{x}_{1,m}, \dots, \mathbf{x}_{N_T,m}]^\top \in \mathbb{C}^{N_T \times L}, \quad (1)$$

where $\mathbf{x}_{n,m} \in \mathbb{C}^{L \times 1}$ denotes the digital samples of the m -th pulse transmitted by the n -th antenna element, L is the code length. Accordingly, the echo of the m -th pulse is given by

$$\mathbf{Y}_{t,m} = \alpha_t e^{j2\pi(m-1)\frac{f_{dt}}{f_r}} \mathbf{a}_R(\theta_t) \mathbf{a}_T^\dagger(\theta_t) \mathbf{X}_m \in \mathbb{C}^{N_R \times L}, \quad (2)$$

where α_t , f_{dt} and θ_t respectively denote the complex amplitude, Doppler frequency and direction of arrival (DOA) of the target and f_r is the pulse repetition frequency (PRF); $(\cdot)^\dagger$ denotes the conjugate transpose. (2) can be equivalently expressed in vectorized form as

$$\mathbf{y}_{t,m} = \text{vec}(\mathbf{Y}_{t,m}) = \alpha_t e^{j2\pi(m-1)\frac{f_{dt}}{f_r}} (\mathbf{I}_L \otimes \mathbf{S}(\theta_t)) \mathbf{x}_m, \quad (3)$$

where $\mathbf{S}(\theta_t) = \mathbf{a}_R(\theta_t) \mathbf{a}_T^\dagger(\theta_t)$ and $\mathbf{x}_m = \text{vec}(\mathbf{X}_m)$.

By stacking all transmit signals of M pulses in a coherent processing interval (CPI), we define $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_M] \in \mathbb{C}^{N_T L \times M}$, and by stacking the received signals, we obtain the overall received vector from the target direction as $\mathbf{Y}_t = [\mathbf{y}_{t,1}, \dots, \mathbf{y}_{t,M}] \in \mathbb{C}^{N_R L \times M}$, which can be expressed as

$$\mathbf{Y}_t = \alpha_t (\mathbf{I}_L \otimes \mathbf{S}(\theta_t)) \mathbf{X} \mathbf{D}(f_{dt}), \quad (4)$$

which can also be vectorized as follows

$$\mathbf{y}_t = \text{vec}(\mathbf{Y}_t) = \alpha_t (\mathbf{D}(f_{dt}) \otimes \mathbf{I}_L \otimes \mathbf{S}(\theta_t)) \mathbf{x}, \quad (5)$$

where $\mathbf{D}(f_{dt}) = \text{Diag}([1, \dots, e^{j2\pi(M-1)\frac{f_{dt}}{f_r}}])$, $\mathbf{x} = \text{vec}(\mathbf{X})$.

Similarly, the clutter model can be formulated as follows

$$\mathbf{y}_c = \sum_{l=-N_a}^{N_a} \sum_{p=1}^{N_c} \alpha_{c,l,p} (\mathbf{D}(f_{c,l,p}) \otimes \mathbf{J}_{-l} \otimes \mathbf{S}(\theta_{c,l,p})) \mathbf{x}, \quad (6)$$

where $\alpha_{c,l,p}$, $f_{c,l,p}$, $\theta_{c,l,p}$ respectively denote the complex amplitude, Doppler frequency and DOA of the p -th clutter patch in the $(r+l)$ -th range cell; $\mathbf{S}(\theta_{c,l,p}) = \mathbf{a}_R(\theta_{c,l,p}) \mathbf{a}_T^\dagger(\theta_{c,l,p})$; N_a denotes the number of adjacent range cells considered, and N_c represents the number of clutter patches within a single iso-range ring. The shift matrix $\mathbf{J}_l = \mathbf{J}_{-l}^\top$ is defined as

$$\mathbf{J}_l(i, j) = \begin{cases} 1, & \text{if } i - j + l = 0, \\ 0, & \text{if } i - j + l \neq 0. \end{cases} \quad (7)$$

We adopt the signal-to-clutter-plus-noise ratio (SCNR) as the metric for sensing performance, which can be expressed as follows

$$\text{SCNR}(\mathbf{w}, \mathbf{x}) = \frac{|\mathbf{w}^\dagger (\mathbf{D}(f_d) \otimes \mathbf{I}_L \otimes \mathbf{S}(\theta_t)) \mathbf{x}|^2}{\mathbf{w}^\dagger (\mathbf{R}_c(\mathbf{x}) + \mathbf{R}_n) \mathbf{w}}, \quad (8)$$

where $\mathbf{w} \in \mathbb{C}^{N_R M L \times 1}$ denotes the receive filter, $\mathbf{R}_c(\mathbf{x})$ represents the clutter covariance matrix, expressed as follows

$$\mathbf{R}_c(\mathbf{x}) = \sum_{l=-N_a}^{N_a} \sum_{p=1}^{N_c} \sigma_{c,l,p}^2 (\mathbf{D}(f_{c,l,p}) \otimes \mathbf{J}_{-l} \otimes \mathbf{S}(\theta_{c,l,p})) \mathbf{x} \mathbf{x}^\dagger (\mathbf{D}^\dagger(f_{c,l,p}) \otimes \mathbf{J}_l \otimes \mathbf{S}^\dagger(\theta_{c,l,p})), \quad (9)$$

where $\sigma_{c,l,p}^2 = \mathbb{E}\{|\alpha_{c,l,p}|^2\}$ denotes the power of the (l, p) -th clutter patch.

B. Communication Model

On the communication side, we consider a multi-user downlink MIMO channel model. Let $\mathbf{H}_k^c \in \mathbb{C}^{M L \times N_T M L}$ denote the communication channel and $\mathbf{c}_k \in \mathbb{C}^{M L \times 1}$ denote the desired symbol vectors of the k -th user. We adopt the multi-user interference (MUI) as the performance metric for the communication subsystem, which is defined as

$$\text{MUI} = \|\mathbf{H}_k^c \mathbf{x} - \mathbf{c}_k\|_2^2. \quad (10)$$

By constraining the MUI energy below a threshold, we ensure that the received signals closely match the intended symbols.

III. PROBLEM FORMULATION

In this section, we formulate the optimization problem for the joint design of the transmit waveforms and receive filters for the DFRC system, as given in (11). As the objective function, we aim to maximize the output SCNR at the receiver. Regarding the constraints, the MUI energy is constrained below a threshold to ensure quality, corresponding to (11a) and transmit beamforming is applied to enhance energy focusing, corresponding to (11b). Moreover, to improve robustness against repeater jamming and fully exploit the benefits of pulse agility, the similarity between the transmit waveforms of adjacent Q pulses is limited, corresponding to (11c). Finally, to ensure hardware-friendly waveforms, peak-to-average power ratio (PAPR) constraints are imposed, corresponding to (11d).

$$\max_{\mathbf{w}, \mathbf{x}} \frac{|\mathbf{w}^\dagger (\mathbf{D}(f_{dt}) \otimes \mathbf{I}_L \otimes \mathbf{S}(\theta_t)) \mathbf{x}|^2}{\mathbf{w}^\dagger (\mathbf{R}_c(\mathbf{x}) + \mathbf{R}_n) \mathbf{w}}, \quad (11)$$

$$\text{s.t.} \max_k \|\mathbf{H}_k^c \mathbf{x} - \mathbf{c}_k\|_2^2 \leq \epsilon_k, \quad \forall k \in \mathcal{K}, \quad (11a)$$

$$\frac{\max_{\theta_s \in \Theta_s} \mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_s) \mathbf{A}_T(\theta_s) \mathbf{x}}{\mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_t) \mathbf{A}_T(\theta_t) \mathbf{x}} \leq \Gamma, \quad (11b)$$

$$\left| \sum_{q=1}^Q \sum_{m=1}^{M-q} \mathbf{x}^\dagger \mathbf{E}_m \mathbf{E}_{m+q} \mathbf{x} \right|^2 \leq \gamma, \quad (11c)$$

$$\text{PAPR}(\mathbf{x}_m) \leq \chi, \quad \forall m \in \mathcal{M}. \quad (11d)$$

In (11), ϵ_k denotes the maximum allowable MUI energy for the k -th user; $\mathbf{A}_T(\theta) = \mathbf{I}_{ML} \otimes \mathbf{a}_T^\dagger(\theta)$, Θ_s represents the set of sidelobe angles, and Γ is the threshold constraining the ratio of the maximum sidelobe level to the mainlobe level. $\mathbf{E}_i = \mathbf{e}_i \otimes \mathbf{I}_{N_T L}$, where $\mathbf{e}_i \in \mathbb{R}^{1 \times M}$ is a row vector of zeros except that the i -th element is 1, and γ denotes the threshold for inter-pulse waveform similarity. χ specifies the threshold constraining the PAPR of the transmit waveforms. Finally, $\mathcal{K} = \{1, 2, \dots, K\}$ and $\mathcal{M} = \{1, 2, \dots, M\}$.

We first employ the cyclic optimization [21] to decompose (11). For a fixed $\mathbf{x}$, the subproblem of $\mathbf{w}$ is given by

$$\max_{\mathbf{w}} \frac{|\mathbf{w}^\dagger (\mathbf{D}(f_{dt}) \otimes \mathbf{I}_L \otimes \mathbf{S}(\theta_t)) \mathbf{x}|^2}{\mathbf{w}^\dagger (\mathbf{R}_c(\mathbf{x}) + \mathbf{R}_n) \mathbf{w}}. \quad (12)$$

This problem can be solved using the Lagrange multiplier method, which admits a close form optimal solution as follows

$$\mathbf{w} = \mu (\mathbf{R}_c(\mathbf{x}) + \mathbf{R}_n)^{-1} (\mathbf{D}(f_d) \otimes \mathbf{I}_L \otimes \mathbf{S}(\theta_t)) \mathbf{x}, \quad (13)$$

where μ is a scalar that does not affect the objective value.

When $\mathbf{w}$ is fixed, the problem associated with $\mathbf{x}$ can be formulated by

$$\max_{\mathbf{x}} \frac{|\mathbf{w}^\dagger (\mathbf{D}(f_{dt}) \otimes \mathbf{I}_L \otimes \mathbf{S}(\theta_t)) \mathbf{x}|^2}{\mathbf{w}^\dagger (\mathbf{R}_c(\mathbf{x}) + \mathbf{R}_n) \mathbf{w}} \quad \text{s.t. (11a)–(11d)}. \quad (14)$$

For the objective function, it can be reformulated into the standard generalized Rayleigh quotient with respect to $\mathbf{x}$ as $\frac{\mathbf{x}^\dagger \tilde{\mathbf{R}}_t \mathbf{x}}{\mathbf{x}^\dagger \tilde{\mathbf{R}}_{cn} \mathbf{x}}$, where $\tilde{\mathbf{R}}_t = \mathbf{S}_t^\dagger \mathbf{w} \mathbf{w}^\dagger \mathbf{S}_t$,

$$\tilde{\mathbf{R}}_{cn} = \sum_{l=-N_a}^{N_a} \sum_{p=1}^{N_p} \sigma_{c,l,p}^2 \mathbf{S}_{c,l,p}^\dagger \mathbf{w} \mathbf{w}^\dagger \mathbf{S}_{c,l,p} + \frac{\mathbf{w}^\dagger \mathbf{R}_n \mathbf{w}}{e_T} \cdot \mathbf{I}_{N_T M L},$$

with $\mathbf{S}_t = \mathbf{D}(f_{dt}) \otimes \mathbf{I}_L \otimes \mathbf{S}(\theta_t)$ and $\mathbf{S}_{c,l,p} = \mathbf{D}(f_{c,l,p}) \otimes \mathbf{J}_{-l} \otimes \mathbf{S}(\theta_{c,l,p})$. To address the resulting fractional optimization problem, auxiliary variables α and β are introduced to construct exponential bounds on the quadratic forms, i.e., $\mathbf{x}^\dagger \tilde{\mathbf{R}}_t \mathbf{x} \geq e^\alpha$ and $\mathbf{x}^\dagger \tilde{\mathbf{R}}_{cn} \mathbf{x} \leq e^\beta$. Additional variables r_t and r_{cn} are further introduced to decouple the inequality constraints. Owing to the monotonicity of the exponential function, minimizing e^α/e^β is equivalent to minimizing $\alpha - \beta$. Similarly, auxiliary variables are introduced to decouple constraints (11a)–(11c). As a result, the problem can be reformulated as

$$\begin{aligned} & \max_{\mathbf{x}, \alpha, \beta, \{\mathbf{e}_k\}, \mathbf{r}_t, \mathbf{r}_{cn}, \{u_s\}, v, g, \varrho_t, \varrho_u} \alpha - \beta \\ & \text{s.t. } \|\mathbf{r}_t\|_2^2 \geq e^\alpha, \quad \mathbf{r}_t = (\tilde{\mathbf{R}}_t)^{\frac{1}{2}} \mathbf{x}, \\ & \quad \|\mathbf{r}_{cn}\|_2^2 \leq e^\beta, \quad \mathbf{r}_{cn} = (\tilde{\mathbf{R}}_{cn})^{\frac{1}{2}} \mathbf{x}, \\ & \quad \|\mathbf{e}_k\|_2^2 \leq \epsilon_k, \quad \mathbf{e}_k = \mathbf{H}_k^c \mathbf{x} - \mathbf{c}_k, \quad \forall k \in \mathcal{K}, \\ & \quad v \geq \varrho_t, \quad v = \mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_t) \mathbf{A}_T(\theta_t) \mathbf{x}, \\ & \quad u_s \leq \varrho_u, \quad u_s = \mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_s) \mathbf{A}_T(\theta_s) \mathbf{x}, \quad \forall s \in \mathcal{S}, \\ & \quad |g|^2 \leq \gamma, \quad g = \sum_q \sum_m \mathbf{x}^\dagger \mathbf{E}_m \mathbf{E}_{m+q} \mathbf{x}, \\ & \quad \frac{\varrho_t}{\varrho_u} \geq \varrho, \quad \text{PAPR}(\mathbf{x}_m) \leq \chi, \quad \forall m \in \mathcal{M}, \end{aligned} \quad (16)$$

where $\mathcal{S} = [1, 2, \dots, S]$, with S denoting the number of sampling points within the sidelobe region Θ_s ; $(\cdot)^{1/2}$ represents the square root of a matrix.

The constraint $\varrho_t/\varrho_u \geq \varrho$ can be removed by exploiting the problem structure. For any optimal solution $(\tilde{\varrho}_t, \tilde{\varrho}_u)$, setting $\varrho_t = \varrho \tilde{\varrho}_u$ yields an equally optimal solution, since $\tilde{\varrho}_t \geq \varrho_t$ and $v \geq \tilde{\varrho}_t \geq \varrho_t$, leaving the objective in (16) unchanged. Hence, we can fix $\varrho_t = \varrho \varrho_u$ and eliminate one variable. Moreover, from $\varrho \varrho_u \leq v \leq \lambda_{\max}(\mathbf{A}_T(\theta_t) \mathbf{A}_T^\dagger(\theta_t)) e_T$, the feasible set of ϱ_u is given by $0 \leq \varrho_u \leq \varrho_{\max}$, where $\varrho_{\max} = \lambda_{\max}(\mathbf{A}_T(\theta_t) \mathbf{A}_T^\dagger(\theta_t)) e_T / \varrho$.

Since the variables v , u_s and ϱ_u remain coupled in the constraint set of (16), the problem is still difficult to solve directly. Therefore, we introduce two additional auxiliary variables ζ_s and ζ_v , leading to the final problem as follows

$$\begin{aligned} & \max_{\mathbf{x}, \alpha, \beta, \{\mathbf{e}_k\}, \mathbf{r}_t, \mathbf{r}_{cn}, \{u_s\}, v, g, \varrho_u} \alpha - \beta \\ & \text{s.t. } \|\mathbf{r}_t\|_2^2 \geq e^\alpha, \quad \mathbf{r}_t = (\tilde{\mathbf{R}}_t)^{\frac{1}{2}} \mathbf{x}, \\ & \quad \|\mathbf{r}_{cn}\|_2^2 \leq e^\beta, \quad \mathbf{r}_{cn} = (\tilde{\mathbf{R}}_{cn})^{\frac{1}{2}} \mathbf{x}, \\ & \quad \|\mathbf{e}_k\|_2^2 \leq \epsilon_k, \quad \mathbf{e}_k = \mathbf{H}_k^c \mathbf{x} - \mathbf{c}_k, \quad \forall k \in \mathcal{K}, \\ & \quad v \geq \zeta_v, \quad v = \mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_t) \mathbf{A}_T(\theta_t) \mathbf{x}, \\ & \quad u_s \leq \zeta_s, \quad u_s = \mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_s) \mathbf{A}_T(\theta_s) \mathbf{x}, \quad \forall s \in \mathcal{S}, \\ & \quad \zeta_v = \varrho \varrho_u, \quad \zeta_s = \varrho_u, \quad 0 \leq \varrho_u \leq \varrho_{\max}, \\ & \quad |g|^2 \leq \gamma, \quad g = \sum_q \sum_m \mathbf{x}^\dagger \mathbf{E}_m \mathbf{E}_{m+q} \mathbf{x}, \\ & \quad \text{PAPR}(\mathbf{x}_m) \leq \chi, \quad \forall m \in \mathcal{M}. \end{aligned} \quad (17)$$

Next, we employ the alternating direction method of multipliers (ADMM) [22] to solve the optimization problem in (17). The augmented Lagrange function of (17) is presented in (15) at the bottom of this page, where $\{\rho_1, \dots, \rho_8\}$ are the penalty parameter, $\{\mathbf{d}_t, \mathbf{d}_{cn}, \{\mathbf{p}_k\}, \lambda, \{\Upsilon_s\}, \{\kappa_s\}, \kappa_v, \xi\}$ are the Lagrange multipliers corresponding to the equality constraints $\mathbf{r}_t = (\tilde{\mathbf{R}}_t)^{\frac{1}{2}} \mathbf{x}$, $\mathbf{r}_{cn} = (\tilde{\mathbf{R}}_{cn})^{\frac{1}{2}} \mathbf{x}$, $\mathbf{e}_k = \mathbf{H}_k^c \mathbf{x} - \mathbf{c}_k$, $v = \mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_t) \mathbf{A}_T(\theta_t) \mathbf{x}$, $u_s = \mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_s) \mathbf{A}_T(\theta_s) \mathbf{x}$, $\zeta_v = \varrho \varrho_u$, $\zeta_s = \varrho_u$ and $g = \sum_q \sum_m \mathbf{x}^\dagger \mathbf{E}_m \mathbf{E}_{m+q} \mathbf{x}$.

In each ADMM iteration, thanks to its decomposition capability, the original problem can be separated into several subproblems with respect to different variables. Due to space limitations, we do not present the detailed formulations and derivations of these subproblems here. The solution methods for the involved subproblems can be referred to in [23], [24].

$$\begin{aligned} \mathcal{L}(\mathbf{x}, \alpha, \beta, \{\mathbf{e}_k\}, \mathbf{r}_t, \mathbf{r}_{cn}, \{u_s\}, v, g, \varrho_u, \varepsilon_t, \varepsilon_{cn}, \{\mathbf{p}_k\}, q, \{\Upsilon_s\}, \kappa_s, \kappa_v, \xi) = & \beta - \alpha \\ & \frac{\rho_1}{2} (\|\mathbf{r}_t - (\tilde{\mathbf{R}}_t)^{\frac{1}{2}} \mathbf{x} + \mathbf{d}_t\|_2^2 - \|\mathbf{d}_t\|_2^2) + \frac{\rho_2}{2} (\|\mathbf{r}_{cn} - (\tilde{\mathbf{R}}_{cn})^{\frac{1}{2}} \mathbf{x} + \mathbf{d}_{cn}\|_2^2 - \|\mathbf{d}_{cn}\|_2^2) + \frac{\rho_3}{2} (|v - \mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_t) \mathbf{A}_T(\theta_t) \mathbf{x} + \lambda|^2 - |\lambda|^2) \\ & + \frac{\rho_4}{2} \left[\sum_{s=1}^S (|u_s - \mathbf{x}^\dagger \mathbf{A}_T^\dagger(\theta_s) \mathbf{A}_T(\theta_s) \mathbf{x} + \Upsilon_s|^2 - |\Upsilon_s|^2) \right] + \frac{\rho_5}{2} \left[\sum_{k=1}^K (\|\mathbf{e}_k - \mathbf{H}_k^c \mathbf{x} + \mathbf{c}_k + \mathbf{p}_k\|_2^2 - \|\mathbf{p}_k\|_2^2) \right] \\ & + \frac{\rho_6}{2} \left(|g - \sum_{q=1}^Q \sum_{m=1}^{M-q} \mathbf{x}^\dagger \mathbf{E}_m \mathbf{E}_{m+q} \mathbf{x} + \xi|^2 - |\xi|^2 \right) + \frac{\rho_7}{2} (|\zeta_v - \varrho \varrho_u + \kappa_v|^2 - |\kappa_v|^2) + \frac{\rho_8}{2} \left[\sum_{s=1}^S (|\zeta_s - \varrho_u + \kappa_s|^2 - |\kappa_s|^2) \right]. \end{aligned} \quad (15)$$

IV. NUMERICAL RESULTS

In this section, we present numerical examples to evaluate the proposed method. An airborne DFRC system is considered at an altitude of 8000 m and velocity 150 m/s, equipped with $N_T = N_R = 10$ antennas. Each CPI contains $M = 12$ pulses with a PRF of 2000 Hz. The receiving azimuth sector $[0^\circ, 180^\circ]$ is divided into $N_p = 181$ clutter patches with power $\sigma_{c,l,p}^2 = 1$, where $l = -1, 0, 1$ denotes the adjacent range cells. The transmit azimuth sector is also $[0^\circ, 180^\circ]$, with sidelobe regions $[0^\circ, 75^\circ] \cup [105^\circ, 180^\circ]$ sampled every 5° . The target normalized spatial and Doppler frequencies are set to 0 and 0.5, respectively, and the transmit energy is $e_T = 1$. We consider $K = 2$ users employing quadrature phase shift keying modulation with energy e_T/K and code length $L = 5$. Communication channels are assumed constant within each CPI and modeled as $\mathbf{H}_k^c = \mathbf{I}_{ML} \otimes \mathbf{h}_k^c$, where $\mathbf{h}_k^c$ is the spatial channel vector. The thresholds in the optimization are set as $\epsilon_k = 1 \times 10^{-3}$, $\Gamma = 5 \times 10^{-2}$, $\gamma = 5 \times 10^{-2}$, and $\chi = 2$.

We first show the receive beampatterns for the proposed joint transmit-receive beamforming scheme and, for comparison, the SCNR-maximization-only scenario [20], as illustrated in Figs. 1(a) and 1(b), respectively.

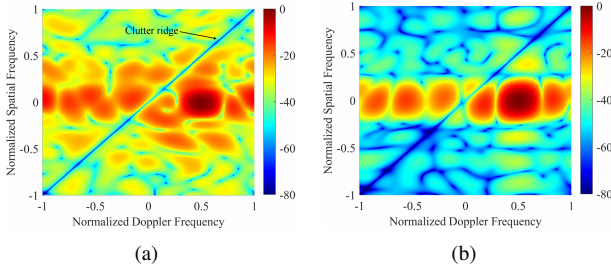


Fig. 1. Synthesized receive beampatterns in the spatial-Doppler domain. (a) Receive beamforming only as in [20]; (b) Proposed method.

From Fig. 1(b), we can see that the designed filters place deep nulls in clutter-dominated regions while maintaining high gain at the target, showing that the filters can coherently combine returns across pulses despite the pulse-agile waveforms.

Comparing Figs. 1(a) and 1(b), both achieve clutter nulling and target gain, but the latter exhibits much higher sidelobes and a more distorted mainlobe. This is because maximizing SCNR alone, although involving waveform design, does not guarantee accurate mainlobe steering or low sidelobe levels in the transmit beampattern, as illustrated in Fig. 2.

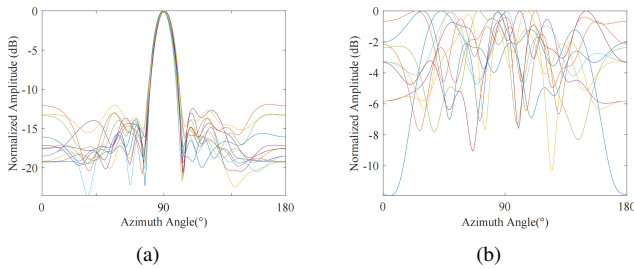


Fig. 2. Synthesized transmit beampatterns. (a) Transmit-receive beamforming. (b) Receive beamforming only.

From Fig. 2, we can observe that employing transmit-receive beamforming ensures that the transmit beampatterns across all pulses maintain stable mainlobe steering and low sidelobe levels. In contrast, using SCNR maximization alone fails to produce stable beampatterns, resulting in suboptimal mainlobe directions and elevated sidelobe levels. This causes energy defocusing, which is manifested at the receiver as mainlobe distortion and increased sidelobe levels in the receive beampattern as shown in Fig. 1.

Finally, we present the performance evaluation of the communication functionality in Fig. 3.

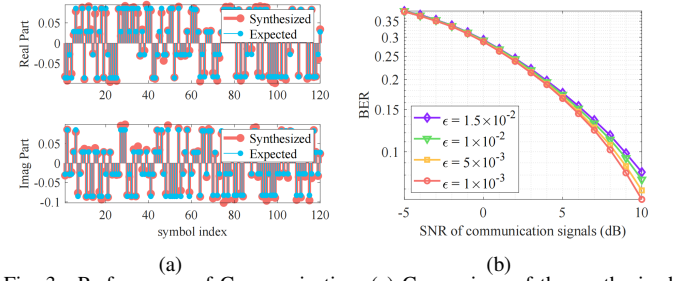


Fig. 3. Performance of Communication. (a) Comparison of the synthesized communication signals with the expected ones. (b) The BER versus SNR of communication signals.

As shown in Fig. 3(a), the received signals at the users align well with the intended transmitted symbols. Moreover, the bit error rate results in Fig. 3(b) demonstrate that the proposed method achieves favorable error-rate performance, indicating that the communication requirements are well satisfied.

Finally, we present the autocorrelation results of the optimized waveforms, with conventional pulse-repetitive waveforms provided for comparison in Fig. 4. It can be observed that the optimized waveforms exhibit significantly lower sidelobe peak levels and fewer sidelobes, indicating improved resilience against repeater jamming, such as interrupted sampling repeater jamming.

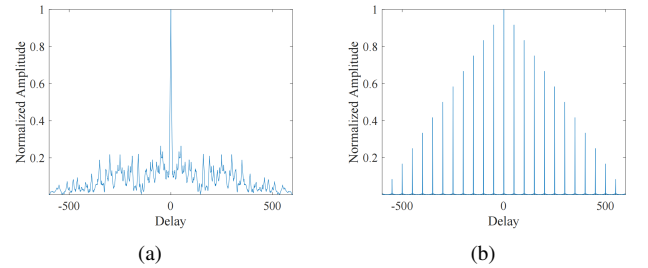


Fig. 4. Autocorrelation results. (a) The optimized waveforms. (b) Conventional pulse-repetitive STAP waveforms.

V. CONCLUSION

This paper investigates the joint optimization of transmit waveforms and receive filters for airborne DFRC systems, aiming to overcome the communication-rate limitations imposed by the pulse-repetitive operation of STAP, while maintaining robust clutter suppression and reliable target sensing, and further enhancing jamming resilience. Overall, this work provides a promising technical foundation for the development of future airborne DFRC systems.

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